



# Pornography diversity: A latent class analysis of pornography users in committed relationships

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## ABSTRACT

While an increasing amount of scholarship has focused on the relationships between pornography use and well-being, most of this research has conceptualized and measured the frequency of pornography use as a single continuum ranging from no use to compulsive use without considering how other contextual factors may impact patterns of use and associations with well-being. Using a sample of 3710 individuals in committed relationships, we explored how various components of pornography use and attitudes toward pornography might be used to create unique profiles of pornography users. We found evidence of six different profiles of users, primarily distinguished based on the frequency of pornography use across varying content types and the configuration of pornography use within their romantic partnership. An exploration of demographic differences between profiles suggested that those who generally abstained from pornography were often older and in longer relationships than those in other profiles. In addition, those who abstained from pornography use both individually and with their romantic partner generally reported the best individual and relational well-being after controlling for a wide range of potentially confounding factors. Those in the Partner Use profile, a group characterized by individuals who generally avoided pornography but had romantic partners who regularly used pornography, were the most likely to report the lowest individual well-being and relationship quality.

Pornography use is an almost ubiquitous part of many lives and is being increasingly accessed and viewed across most demographic segments of the population (Carroll et al., 2008; Mestre-Bach, Blycker, & Potenza, 2020). In addition to pornography use being increasingly common, people's attitudes toward pornography use are also becoming more accepting, suggesting that pornography use is now a normative aspect of both individual and couple sexuality (Carroll et al., 2008; Price, Patterson, Regnerus, & Walley, 2016; Willoughby, Rhoades, & Carroll, 2021). Due to this normative element of pornography use, it is crucial for scholars to understand behavioral trends and attitudes related to how both individuals and couples decide to engage with sexually explicit content. While a growing amount of scholarship has examined pornography's potential effect on both individuals (Hald & Malamuth, 2008; Miller, Hald, & Kidd, 2018) and on couples (Daneback et al., 2009; Poulsen et al., 2013; Willoughby, Carroll, Busby, & Brown, 2016), less research has sought to explore crucial behavioral patterns that are vital to providing context and nuance to these findings.

Scholars have repeatedly noted many methodological flaws and limitations of the current research exploring pornography use (see

Campbell & Kohut, 2017; Willoughby, Leonhardt, & Augustus, 2020). Among the most cited issues brought up by scholars are limited one-item measurement (Busby, Willoughby, Chiu, & Olsen, 2020), a lack of measurement for the content being viewed (Willoughby & Dover, 2022), overlooking how masturbation intersects with pornography use (Kohut et al., 2020), and not considering important contextual factors such as how and if couples use pornography together (Willoughby et al., 2020). While previous scholarship has explored specific unique elements of pornography viewing individually, such as dyadic use patterns (Carroll, Busby, Willoughby, & Brown, 2017), the role of pornography genre (Hald et al., 2018) and perceived realism of pornography (Wright, Herbenick, & Paul, 2022), little research has sought to explore many of these factors concurrently to attempt a more comprehensive exploration of pornography viewing habits. Due to this lack of research tying these contextual factors together, scholars still having limited information regarding if pornography viewing patterns are unique or follow similar or dissimilar patterns between people when considering a range of factors regarding how individuals consume sexually explicit material. Current research largely assumes that pornography use exists on a

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continuum, ranging from abstainers who avoid pornography to those who watch pornography regularly or may experience problematic or compulsive pornography use. But this simple continuum of individual use ignores factors that researchers have noted as being critical to understanding the potential correlates of pornography use. These variations may be particularly relevant in the relational arena where varying patterns of both individual and couple pornography use may be one of the reasons outcome research in this area has produced a mixture of results which are difficult to interpret (Willoughby et al., 2020).

To date, limited research has found that pornography use does vary across a variety of factors, including usage variations by gender (Carroll et al., 2017; Vanden Abeele, Campbell, Eggermont, & Roe, 2014; Willoughby, Rhoades, & Carroll, 2021), motivational patterns (Paul & Shim, 2008), culture (Rowland & Uribe, 2020), and between couples (Willoughby et al., 2020). These variations may have implications as individuals form and maintain committed relationships in a modern context where pornography use is increasingly normative. However, no study to date has been able to utilize large scale sampling with detailed measurement on pornography use to explore common pornography viewing trends and patterns for those within romantic relationships. To that end, the current study utilized a latent profile analysis of a large national sample of adults in committed relationships in the United States to explore common trends in pornography viewing patterns when considering several factors related to viewing and how these patterns are related to background factors, individual well-being, and relationship quality.

## 1. Pornography use

Pornography use is an increasingly common part of many lives (Carroll et al., 2008; Mestre-Bach et al., 2020; Price et al., 2016). Price et al. (2016) found that pornography use has increased for both men and women since the 1970's to the 2000's. In a sample of men, Carvalheira, Træen, and Stulhofer (2015) found that for men who report masturbating at least once per week, 70% of them reported a similar frequency of using pornography. More recently, Mestre-Bach et al. (2020) noted that pornography use increased during the COVID-19 pandemic. Trends are similar when examining attitudes toward pornography use (O'Reilly et al., 2007; Price et al., 2016). Among a sample of college students, O'Reilly et al. (2007) found that the majority of both men and women say that pornography is "ok", with men being significantly more likely to report this than women. In addition to the research on pornography use by individuals, pornography use by couples is also a common occurrence and many individuals in committed relationship view pornography both alone and with their partner (Carroll et al., 2016; Willoughby & Leonhardt, 2018). One study suggested that about 80% of males in relationships who used pornography viewed it alone the majority of the time, and about 68% of females in relationships reported a similar pattern (Willoughby & Leonhardt, 2018). Although these findings highlight how couple pornography use is a common pattern in relationships, it is also clear that pornography use alone is still the dominant trend among adults.

As pornography use has become more normative and frequent, scholars have increasingly been interested in the correlates of this use in a variety of contexts. The correlates of pornography have been widely studied and examined by scholars in recent years (Daneback et al., 2009; Hald & Malamuth, 2008; Miller et al., 2018; Poulsen, Busby, & Galovan, 2013; Willoughby et al., 2016). Hald and Malamuth (2008) found that both men and women perceived positive effects of pornography consumption on a variety of factors including their sex life, attitudes toward sex, sexual knowledge, and life in general. Miller et al. (2018) sampled a group of heterosexual men who have viewed pornography and found that these men reported a greater positive than negative impact in their lives. Clearly, as these studies show, self-reported effects of pornography use can often be positive, especially among men.

Previous research has also suggested a variety of negative outcomes

that appear to occur with pornography use, especially in a relational context. Some negative outcomes for relationships include links with more aggression, less relationship satisfaction, lower sexual quality, and a higher likelihood of divorce (Daneback et al., 2009; Perry & Schleifer, 2018; Poulsen et al., 2013; Sun, Bridges, Johnson, & Ezzell, 2016; Willoughby et al., 2016; Willoughby & Leonhardt, 2018; Wright, Tokunaga, Kraus, & Klann, 2017). Willoughby et al. (2016) observed that pornography use was associated with less positive communication, more relational aggression, less relationship satisfaction, and less stability in committed relationships. Other research has also found that pornography use is associated with lower relationship satisfaction and relationship adjustment (Muusses, Kerkhof, & Finkenauer, 2015; Willoughby & Leonhardt, 2018; Wright et al., 2017). In addition, studies have shown that pornography use had a negative association with sexual quality and satisfaction for both men and women (Poulsen et al., 2013; Sun et al., 2016). In a more nuanced way, Daneback et al. (2009) found that couples who use pornography have more permissive erotic climates, but that these couples also report more sexual dysfunctions.

Combined, these general findings associated with pornography use suggest that pornography has increasingly become a normal part of both individual and couple sexual intimacy, and that the utilization of pornography may have a varied influence on relationships. Most adult couples are now navigating how pornography use fits within their couple intimacy patterns as they discuss (or avoid) boundaries and behaviors tied to individual and joint pornography use (see Willoughby, Rhoades, & Carroll, 2021). A growing body of research has also suggested that these decisions to engage with or avoid pornography appear to have ramifications for both individual and relational health, with current research showing mixed results but trending toward pornography use generally being associated with negative well-being for most couples. However, this basic connection between pornography use and both individual and relational health is likely much more complex as evidenced by more recent research, which has suggested that associations between pornography use and well-being are likely heavily influenced by a variety of contextual factors (Grubbs, Perry, Wilt, & Reid, 2019; Perry, 2020; Willoughby & Dover, 2022; Willoughby et al., 2020).

### 1.1. Variations in the pattern of pornography use

While pornography use has generally been increasing, significant variations in the utilization of pornography likely exist across individuals. For example, factors such as religiosity and gender likely serve as key factors that may create between-person differences in how pornography is utilized at the individual level (Carroll et al., 2008, 2017; Hardy, Steelman, Coyne, & Ridge, 2013; Patterson & Price, 2012). Gender has been a commonly discussed factor that creates differences in pornography patterns across individuals. Women tend to report less frequent pornography use generally, use pornography alone less often, and have less awareness about the amount of pornography that their partner uses compared to men (Carroll et al., 2008, 2017; Price et al., 2016; Willoughby & Leonhardt, 2018). There have also been studies which examined how pornography use varies by religiosity, with results generally finding a negative association between religiosity and pornography use (Hardy et al., 2013; Patterson & Price, 2012; Wright, Bae, & Funk, 2013). Together these studies suggest that understanding and capturing between-person variations in use may be a key element in providing additional context and nuance to previous research linking pornography use to well-being.

While these pockets of scholarship have hinted at the possibility that variations between individuals in their pornography use likely exist based on demographic characteristics, only a limited number of studies have attempted to tackle this question directly, especially when examining individual details related to the use of pornography. Scholars in previous work have attempted to understand profiles of pornography users (Brown, Conner, & Vennum, 2017; Brown, Durtschi, Carroll, & Willoughby, 2017; Bóthe, Tóth-Király, Potenza, Orosz, & Demetrovics,

2020; Rasmussen & Bierman, 2018; Willoughby, Carroll, Nelson, & Padilla-Walker, 2014; Willoughby, Young-Petersen, & Leonhardt, 2018; Štulhofer, Tafro, & Kohut, 2019), and have provided some insight into the importance of exploring these variations.

In one of the earliest attempts to distinguish group differences based on variations in pornography use, Willoughby et al. (2014) used a sample of emerging adults from the United States to examine common usage patterns using a single-item pornography frequency item and sexual engagement. Using separate analyses for males and females, they found evidence of three unique profiles for each gender. Profile membership was significantly related to both risk-taking behaviors and indicators of mental health, with profiles associated with more pornography use and greater engagement with sexual behaviors more likely to report risk-taking and lower mental health. Brown, Conner, and Vennum (2017) examined latent profiles of pornography users with a sample of undergraduate students from a university ( $n = 635$ ). Using a single-item measure to capture frequency of pornography use, and combining frequency with pornography acceptance, these scholars found that both men and women could be separated into a group of high frequency pornography users who accepted pornography and a group of “dabblers” who used pornography less frequently and had more negative views of pornography. In a follow-up paper Brown, Durtschi and others (2017) used the same one-item scale for frequency of pornography use in another sample of college students but added a measure of motivation to use pornography and found evidence of three profiles of college students who use pornography.

Štulhofer et al. (2019) gathered a sample of adolescents in Croatia ( $n = 1289$ ) and used a single-item measure in their longitudinal latent profile analysis to examine between-person variations in pornography use, finding significant group differences across time in how pornography usage varied over the waves of the study. This longitudinal trajectory approach has also yielded similar results in other studies, where scholars have noted that varying trajectories of pornography use appear when examining the time period from early adolescence through young adulthood (Willoughby et al., 2018). More recently, Bóthe et al. (2020) examined a single item pornography use measure and problematic pornography use across three different studies and a sample of over 14,000 individuals. Three profiles of users were found, with a single low frequency profile and two high frequency profiles, differing on their problematic use scores. These profiles were found to differ on a range of outcomes, including depressive symptoms and self-esteem.

Collectively, these previous studies have provided strong evidence that while pornography use itself likely follows a singular continuum that ranges from low to high frequency use when only considering pornography use frequency, other contextual factors that influence viewing patterns may create behavioral patterns better captured by examining profiles of use that consider multiple factors rather than continuous measures of only frequency. However, while such variations and potential group differences appear meaningful, no consistent typology of pornography users has emerged, likely due to severe methodological limitations of many of these previous studies. These studies have often been narrowly focused on specific contexts (problematic use, the interaction of pornography and sexual behavior, etc.) and have exhibited the same measurement challenges highlighted by several scholars in this area (Busby et al., 2020; Kohut et al., 2020). In almost all the previous cited studies, a narrow, single-item measure of pornography use was utilized to determine profile differences. This has limited the utility of these profiles and likely contributed to these groupings being referred to and applied largely only in these singular studies. Additionally, several critical aspects of how pornography is utilized, especially in a relational context, have yet to be incorporated into these types of studies. For example, no study exploring between-person variations in pornography use has considered dyadic factors, including the individual pornography use of both partners and the pornography use done together as a couple. Given that variations in pornography use for those in committed relationships likely occur between individuals,

between partners, and between couples, ignoring these patterns has only given us a small glimpse into how pornography use may truly vary across individuals in relationships.

The present study addresses many of these previous limitations and is the first to utilize latent profile analysis with a large sample from the United States that includes a content-specific focus on pornography use to understand the profiles of pornography users and outcomes for their relationships. It is also the first to examine how individual, partner, and joint pornography use might combine to create unique variations in pornography use patterns. These additions provide a unique contribution to this area of scholarship and will provide valuable information on how pornography is often utilized for those in romantic partnerships and how these variations in use are related to both individual and relational well-being.

## 1.2. Present study

While previous research has both suggested the importance of exploring between-person differences in patterns of pornography use and suggested that such variations likely exist in meaningful ways (Brown, Conner, & Vennum, 2017; Brown, Durtschi, et al., 2017; Bóthe et al., 2020; Štulhofer et al., 2019), methodological limitations to date have largely kept large-scale explorations of variations in pornography use patterns within committed relationships to a minimum. For the present study, we opted to focus on those individuals who are currently in a committed relationship. While pornography use has important individual and clinical implications (Park et al., 2016), past research has suggested that how individuals and couples navigate personal and relational pornography use is becoming an essential part of modern relationship formation and maintenance that is only weakly understood (Willoughby, Rhoades, & Carroll, 2021). Given that pornography use, both individually and jointly, has consistently been linked to multiple indicators of relational quality (Willoughby & Leonhardt, 2018), such an exploration of common pornography patterns for those also navigating relational dynamics seems especially warranted.

Our choice of outcomes and well-being indicators was guided by previous research. Individual well-being was assessed by both general distress and life satisfaction. General distress was assessed using a validated scale exploring both depression and anxiety symptoms. Pornography use and mental health have long been linked in previous research (Bóthe et al., 2020; Willoughby et al., 2014), with higher pornography use, especially problematic or compulsive use, being associated with higher rates of depressive symptoms (Bóthe et al., 2020). While less studied, pornography use has also been associated with life satisfaction (Hald & Malamuth, 2008) and life satisfaction represented a well-studied and robust indicator of self-perceived well-being (Diener, Lucas, & Oishi, 2002).

For relationship quality, we focused on relationship satisfaction and stability, two well-studied and often utilized markers of relationship quality. While relationship satisfaction has long been linked to both pornography use (Muusses et al., 2015; Willoughby & Leonhardt, 2018; Wright et al., 2017) and healthy couple processes (Willoughby et al., 2016), some scholars have noted the limitations of assessing couple health simply based on self-reported satisfaction or happiness (Fincham, Stanley, & Beach, 2007). Relationship stability then represented an alternative aspect of relational well-being by exploring the underlying commitment tied to the relationship. Relationship stability has also been linked to many elements of relationship health and well-being (Roberson, Norona, Lenger, & Olmstead, 2018) and has been associated with general pornography use in previous studies (Willoughby et al., 2016; Willoughby, Rhoades, & Carroll, 2021).

The first primary focus of the present study was to examine if distinct groups of pornography users could be found by using a variety of measures that capture the complexity of modern pornography use for those in a romantic relationship. We utilized latent profile analysis, a common tool for examining patterns of between-person variation, to

explore patterns across numerous pornography use and attitude indicators. We utilized measures of both individual, perceived partner, and joint pornography use, as previous research has suggested that each of these behaviors represented a unique type of pornography use behavior (Willoughby et al., 2020). In addition, previous scholars have argued that it is vital to distinguish pornography users based on the content of pornography viewed, particularly distinguishing pornography content that is violent or deviant (Kingston, Fedoroff, Firestone, Curry, & Bradford, 2008). For these reasons, we examined each of these three types of pornography use across measures for both general and aggressive forms of pornographic content. To supplement these purely behavioral markers, we also assessed attitudes toward pornography in the form of general acceptance and assessed the average duration of pornography viewing sessions. Finally, we included a measure of masturbation frequency tied to pornography use, as many scholars have also noted that masturbation may be a critical differentiating factor in how pornography influences individual well-being (Miller, McBain, Li, & Raggatt, 2019; Perry, 2020; Wright et al., 2017). Specifically, scholars have noted that whether an individual regularly masturbates while viewing pornography may be a critical confounding factor in understanding links between pornography use and well-being by influencing perceptions of one's partner and romantic relationship (see Wright et al., 2017).

Given the exploratory nature of this analysis and the lack of previous research that has had access to the measurement and large sample needed for this type of analysis, no a priori hypotheses regarding the number or types of profiles generated by this analysis were made. Instead, we were guided by the following broad research question:

**RQ1.** When considering the frequency of pornography use across multiple contexts, the type of pornography consumed, the average duration of pornography use, the acceptance of pornography, and the amount of masturbation that accompanies pornography use, how many unique profiles of pornography users can be distinguished for those currently in a committed relationship?

Once clear profiles of pornography users were found, we next sought to explore differences between these profiles. First, we explored how these profiles might differ based on several demographic and background factors such as age, gender, sexual orientation, relationship status, and religiosity. Next, we explored individual and relational well-being differences across profiles. Here, it is possible to explore generalized hypotheses, as previous research has suggested that more pornography use is generally associated with less individual (Camilleri, Perry, & Sammut, 2021; Willoughby et al., 2014) and relational well-being (Sun et al., 2016; Willoughby et al., 2016; Wright et al., 2017). As such, we explored the following two hypotheses:

**H1.** Profiles that include more frequent pornography use across multiple contexts will be more likely to report lower levels of personal well-being, including more general distress and less life satisfaction.

**H2.** Profiles that include more frequent pornography use across multiple contexts will be more likely to report lower levels of relational quality, including less relationship stability and less relationship satisfaction.

## 2. Methods

### 2.1. Sample

All data collection procedures for this study were approved by the Institutional Review Board at Brigham Young University. Participants for this study were identified and administered an online survey by the online data collection firm, Qualtrics. The sample was recruited from across the United States. Inclusion criteria for the study were that participants were over the age of 18 and currently in a committed, romantic relationship. All participants for this study completed an informed

consent form prior to being asked any questions in the survey. As part of that informed consent, participants were given an overview of the study, assured of the confidentiality of their data, and explained their rights as a research participant. Following completion of the survey, participants were compensated directly by Qualtrics, at the standard rates they utilize for online panels. To substantiate data quality, five attention checks were inserted throughout the survey. Examples of these items were "If you are reading this question, please select 'Somewhat important'" and "If you are reading this question, please move the slider to '65'". Participants who did not pass all attention checks throughout the survey were removed from the final sample. A total of 21 individuals were removed from the data due to bad data related issues or completing the survey in an unreasonably fast amount of time. Those who were removed were not significantly different on biological sex, relationship length, education, or income. Those who were removed were significantly ( $p < .001$ ) older ( $M = 44.75$ ) than those in the final sample ( $M = 39.67$ ). Those who were removed had significantly ( $p < .001$ ) higher average religious attendance ( $M = 2.91$ ) compared to those in the final sample ( $M = 2.51$ ). Those removed also had a significantly ( $p < .001$ ) higher average number of children ( $M = 2.25$ ) compared to those in the final sample ( $M = 1.91$ ). Missing data was very low on all reported variables (2% or less).

### 2.2. Participants

The final sample for this study consisted of 3710 individuals, 71% female and 28% male with 1% identifying as transgender or non-binary. The mean age for the sample was 38 years old ( $SD = 11.19$ ), and had been in their relationships for an average of 8.8 years ( $SD = 9.8$ ). Fifty-five percent of the sample were in their first marriage, 23% were cohabiting, 12% were remarried, 8% were dating and 2% were in an open relationship. Regarding sexual orientation, 72% identified as mostly or completely heterosexual, 17% as mostly or completely homosexual and 11% as bisexual. In terms of education, 77% received some education after high school. Thirty six percent had completed a bachelor's degree or higher. The largest religious denomination was Protestant Christian (31%) followed by No affiliation (25%) and Roman Catholic (18%). Thirty-three percent reported never attending religious services, 22% attend once or twice/year, 15% attend several times/year, 12% attend at least monthly, and 18% attend weekly. The sample was 79% White, 11% Black, 2% Mixed/Other, 4% Asian, 3% Hispanic and 1% Native American.

### 2.3. Measures

Several scales and single-item variables were used to determine latent profiles. These variables capture pornography use across the relationship as well as other behaviors connected to pornography use and attitudes toward pornography.

#### 2.3.1. Pornography use

To capture both the content and the context of mainstream pornography use, the Pornography Usage Measure (PUM) was used (Busby et al., 2020). Participants received the prompt: "Please indicate how often in the last 12 months you have viewed or used the following sexual content". The scale included 3 different iterations, one asking about use alone without a romantic partner, one asking about use with a partner (joint use), and one asking about how much they believed their partner was viewing pornography for each item. The mainstream pornography use measure consisted of 7 different items, with a response scale from 1 ("Never") to 6 ("Every day or almost every day"). Example items included: "An image of a woman alone posing in a suggestive way without any clothes on"; "An image of a heterosexual couple having sex which shows the man's penis penetrating the woman"; "A video showing two naked women or men manually stimulating each other". Higher scores represented higher pornography use. Reliabilities for each scale



were good (alone:  $\alpha = 0.94$ ; partner:  $\alpha = 0.96$ ; joint:  $\alpha = 0.96$ ).

The aggressive pornography use measure (Busby et al., 2020) used the same prompt, scale, and iterations (alone use, joint use, partner use) as the mainstream use, but different items were used to assess using more aggressive forms of pornography. Items included: “a video of a man forcing a woman to have sex against her will”; “a video of a man hitting or slapping a woman while having sex”; “a video of a man choking a woman during sex”. Higher scores reflected higher aggressive pornography use. Cronbach’s alpha for each of the scale iterations were good (alone:  $\alpha = 0.90$ ; partner:  $\alpha = 0.90$ ; joint:  $\alpha = 0.92$ ).

### 2.3.2. Pornography acceptance

The pornography acceptance scale used in the latent profile analysis originated in the RELATE study (Busby, Holman, & Taniguchi, 2001). Participants were given four statements and asked to what extent they agreed on a scale from 1 (“Strongly Disagree”) to 5 (“Strongly Agree”). The four statements had the same structure but asked about four different groups of people: “Viewing pornography is an acceptable way for (teens, adults not in a relationship, adults in a relationship, married adults) to express their sexuality”. Higher scores represented higher acceptance of pornography. Cronbach’s alpha was high for this scale ( $\alpha = .87$ ).

### 2.3.3. Duration of pornography use

Duration of pornography use was measured with one item. Participants were prompted: “Think about your typical pornography viewing behavior. When you watch pornography, how long does this viewing session typically last on average?”. Individuals in the sample then responded on a scale from 1 (“Less than 5 min”) to 6 (“Longer than 2 h”). For those who did not report viewing pornography, their score on this item was set to 0.

### 2.3.4. Masturbation with pornography

Because masturbation has been shown to be strongly linked with pornography use (Miller et al., 2019; Perry, 2020), a measure was included in the latent profile analysis to help capture this behavior. Participants were asked: “On average, what percentage of the time do you masturbate when viewing pornography?”. Respondents used a slider scale from 0 to 100 to record their response.

For outcomes variables, two variables were focused on relationship outcomes (relationship satisfaction and stability), while the other two variables were focused on individual outcomes (general distress and life satisfaction).

### 2.3.5. Relationship satisfaction

The first relationship outcome variable was relationship satisfaction. The relationship satisfaction scale comes from the RELATE assessment (Busby et al., 2001). Participants were asked: “In your relationship, how satisfied are you with the following?”. They then responded to items on a 5-point scale (“Very Dissatisfied” to “Very Satisfied”). Items about the relationship included: “the physical intimacy you experience”; “the love you experience”; “the amount of relationship equality you experience”; “your overall relationship with your partner”. Higher scores represented higher levels of relationship satisfaction. The RELATE satisfaction measures employed in this study have shown high test-retest reliability (between .76 and .78) and validity data have consistently shown that this scale is highly correlated with an existing relationship satisfaction and quality scale (Revised Dyadic Adjustment Scale) in both cross-sectional and longitudinal research (Busby et al., 2001, 2009). The 6-item scale had good reliability in the current sample ( $\alpha = 0.93$ ).

### 2.3.6. Relationship stability

The relationship stability scale also comes from the RELATE study (Busby et al., 2001) and items were adapted from earlier work by Booth, Johnson, and Edwards (1983). The 3-item scale asked to participants included questions about the general stability of the relationship: “How often have you thought your relationship (or marriage) might be in

trouble?”; “How often have you and your partner discussed ending your relationship (or marriage)?”; “How often have you broken up or separated and then gotten back together?”. Participants responded on a scale from 1 (“Never”) to 5 (“Very Often”). All of the items were reverse coded so that higher scores represented higher levels of relationship stability. Previous studies have shown this scale to have test-retest reliability values between .78 and .86, to be appropriately correlated with other relationship quality measures, and to be valid for use in cross-sectional and longitudinal research (Busby et al., 2001, Busby et al., 2020, 2009). The 3-item scale had good reliability in the current sample ( $\alpha = 0.84$ ).

### 2.3.7. General distress

The first individual outcome utilized the MASQ-General Distress scale (Clark & Watson, 1991). Participants were given the prompt: “Below is a list of feelings, sensations, problems, and experiences that people sometimes have”. They then responded on a scale from 1 (“Not at all”) to 5 (“Extremely”) which reflected how much they have felt or experienced the items on the list. Example items included: “felt confused”; “felt inferior to others”; “felt hopeless”; “felt irritable”; “had trouble making decisions”. Higher scores on this scale represented higher levels of general distress. Reliability for the 10-item scale had good reliability ( $\alpha = 0.94$ ).

### 2.3.8. Life satisfaction

The second individual outcome used was a one item measure asking participants how they would rate their overall life satisfaction. They responded on a scale from 1 (“Very dissatisfied”) to 7 (“Very satisfied”). While we used only one item, scholars have noted that one-item assessments of life satisfaction are generally valid and reliable assessments of satisfaction rates and largely equivalent to multi-item assessments (Cheung & Lucas, 2014; Jovanović, 2016).

Demographic characteristics were included in the analysis as covariates for the outcome comparisons between profiles, and were also used to examine differences in profile composition. Gender was coded with 0 = male, 1 = female. Those in the sample who identified as transgender were removed due to their low sample size and to provide a clear comparison of those who identified as male and female. Age was assessed by asking participants to report their current age when they took the survey. Religiosity was broadly assessed by combining three items that examined both intrinsic religiosity and extrinsic religious behavior. One item, “Spirituality is an important part of my life” was assessed on a scale asking how true this statement was (1 = “Never” to 5 = “Often”). A second item asked how often the participant prayed and was measured on the same scale (1 = “Never”; 5 = “Often”). The final item asked, “how important is your religious faith to you?” with responses ranging from 1 (“Not important at all”) to 5 (“Very important”). All items were standardized and averaged to create a composite religiosity scale. The reliability of this measure was acceptable ( $\alpha = 0.84$ ). Participants were given a slider scale and asked to report their sexual orientation. The scale ranged from “Completely Homosexual” to “Completely Heterosexual”. Education was assessed with a single item asking participants to report their highest level of completed education representing sequentially higher levels of education (1 = less than high school; 7 = Advanced degree (JD, Ph.D., MD, etc.)). Marital status was controlled for by coding individuals into two groups based on their status relationship status (married = 1, not married = 0). Relationship length was measured by a single item asking each participant how many months they had been in their current relationship. The number of biological children was assessed by one item. Those who reported more than eight children were collapsed into the highest category (8) to reduce skewness in the variable.

## 2.4. Analysis plan

Prior to all analyses, variables were examined for assumptions of normality and outliers, with no issues being found. To detect profiles of

pornography users across the sample, latent class analyses were run using Mplus software. Latent profile solutions with two through ten profiles were first examined for model fit. Fit statistics were first examined to determine the profile solution with the best fitting statistics, including an examination of BIC, AIC, entropy, the Lo-Mendell-Rubn (LMR) test and Likelihood Ratio Tests. Based on recommendations by Nylund-Gibson and Choi (2018), the best fitting solution was based on models that created with the lowest BIC and AIC scores, entropy values close to 1, and LMR and Likelihood ratio tests that were significant (indicating they were a better fit than the previous n-1 profile solution). Profile solutions were then examined for practical and theoretical relevance to determine the best overall solution based on the data.

Once the optimal number of profiles was determined, individuals in the dataset were assigned a profile based on conditional probabilities, with each person being assigned the profile with the highest probability. These profiles were then examined for broad differences in both background factors and well-being. Differences in categorical variables were examined with chi-square tests while continuous variables were examined with MANOVA models. Here, broad mean differences across all potential dependent variables were first explored. If significant, step-down F-tests were then run to examine where significant mean differences were located. For significant F-tests, post-hoc comparisons between profile groups were run using a Bonferroni adjustment.

### 3. Results

#### 3.1. Latent profile results

Variables included in the latent profile analysis included pornography use alone, joint pornography use with a romantic partner, perceived partner pornography use alone, aggressive pornography use alone, joint aggressive pornography use, perceived partner aggressive pornography use alone, masturbation with pornography frequency, pornography acceptance, and average duration of pornography viewing. Fit statistics for each profile solution can be found in Table 1.

As shown in Table 1, fit statistics generally improved from one to six profiles. The solution for the seventh profile did produce a decrease in AIC and BIC, showed a slight decrease in entropy but, most notably, produced a non-significant LMR test, suggesting that the addition of a seventh profile did not significantly improve model fit. An examination of profile solutions from eight to ten confirmed that additional profiles were only marginally improving model fit. Based on this information, the six-profile solution was examined for practical significance. This solution produced profiles of adequate cell sizes (all  $n > 100$ ) and created theoretically meaningful profiles. For this reason, the six-profile solution was deemed the most appropriate for the data and was utilized in the remaining analyses.

Means across indicator variables for the six profiles are shown in

**Table 1**  
Profile fit statistics for two to 10 profiles.

| Profiles | AIC          | BIC          | Entropy     | LMR test p value | Likelihood Ratio Test p value |
|----------|--------------|--------------|-------------|------------------|-------------------------------|
| 2        | 108843       | 109017       | .993        | <.001            | <.001                         |
| 3        | 103692       | 103929       | .919        | <.001            | <.001                         |
| 4        | 100159       | 100458       | .934        | .003             | <.001                         |
| 5        | 97826        | 98187        | .934        | <.001            | <.001                         |
| 6        | <b>95891</b> | <b>96314</b> | <b>.939</b> | <b>.003</b>      | <b>&lt;.001</b>               |
| 7        | 94006        | 94492        | .938        | .47              | <.001                         |
| 8        | 92915        | 93463        | .892        | .07              | <.001                         |
| 9        | 91709        | 92319        | .896        | .64              | <.001                         |
| 10       | 91233        | 91906        | .897        | .48              | <.001                         |

*Note.* Included variables were pornography use alone, joint pornography use, perceived partner pornography use, aggressive pornography use alone, aggressive joint pornography use, perceived partner aggressive pornography use, pornography acceptance, and masturbation to pornography.

**Table 2.** As can be seen, the two largest profiles found included a profile of *Abstainers* ( $n = 1,796$ , 47.9% of the sample) and a profile of what were labeled *Solo Users* ( $n = 1,264$ , 33.7%). Those in the Abstainer profile generally reported the least engagement with pornography across all pornography use variables and had the lowest acceptance of pornography. While those in this profile did not completely abstain from any form of pornography use, they generally reported low frequency use of pornography. The Solo User profile was distinguished from this Abstainer profile by their moderate use of both pornography and aggressive pornography alone in the absence of regular pornography use with a partner (or by their partner). They had generally moderate acceptance of pornography and reported the highest average masturbation frequency to pornography viewing, suggesting a behavioral pattern of solo pornography use primarily for the purpose of masturbation.

The remaining four profiles were markedly smaller. One group was labeled *Partner Porn Users* ( $n = 152$ , 4.1%) and were distinguished by other groups by their high reported perceived partner pornography use (both general and aggressive). They tended to have low frequencies of both personal and partner pornography use and held moderate attitudes toward pornography. The final three profiles were comprised of individuals who had a similar pattern of responses, but with increasing frequency of pornography use. *Moderate Users* ( $n = 272$ , 7.3%) generally reported regular pornography use at the individual level and with one's partner while also reporting consistent partner pornography use. Each of these types of pornography use were reported, on average, to take place roughly a few times per month. They held fairly accepting views of pornography. The next profile, labeled as *Strong Users* ( $n = 145$ , 3.9%), reported a very similar pattern of responses but with pornography usages that were higher, taking place weekly across most variables on average. The final profile, *Heavy Users*, ( $n = 121$ , 3.2%), reported pornography use individually and with one's partner at very high frequencies, typically averaging several times per week. This profile also reported the highest average pornography viewing duration and the highest acceptance of pornography.

#### 3.2. Gender sensitivity analysis

Given significant differences in pornography viewing patterns established in previous scholarship, we next sought to examine if running the latent class models across genders masked significant underlying differences in profiles based on gender. To conduct this sensitivity analysis, the sample was split between men and women and the latent profile analysis was rerun separately for each gender. This sensitivity analysis largely confirmed that our six profile model was replicated across both men and women with two slight differences.<sup>1</sup> For the model of just men, the six-profile model was deemed the best fitting model using the same criteria as the model for the entire sample (AIC, BIC, entropy, LMR test). This six-profile solution mirrored the results of the full sample with one exception. For men, the Partner User profile was not found. Instead, a sixth profile was found that was comprised of men who used mainstream pornography on a regular basis but did not regularly use pornography with aggressive content. All other profiles were similar to those found in the full sample.

For women, the six-profile model was also deemed the best fit based on all criteria used, although this six-profile model had poorer fit than that model for the full sample and the sample of just men. Like men, an examination of these six profiles suggested that they mirrored those found in the full sample with one exception. For women, the Solo User profile was not found. Instead, a profile of women who used only mainstream pornography alone and with a partner but in the absence of regular engagement with aggressive pornography was found.

Given that sensitivity analyses largely confirmed the presence of the

<sup>1</sup> Full results available from the first author upon request.

**Table 2**

Means on indicator variables for six profile solution of pornography users.

| Variable                              | Partner Users | Solo Users | Abstainers | Moderate Users | Strong Users | Heavy Users |
|---------------------------------------|---------------|------------|------------|----------------|--------------|-------------|
| Porn Use Alone                        | 1.88          | 2.31       | 1.20       | 2.59           | 3.54         | 4.80        |
| Joint Porn Use                        | 1.33          | 1.65       | 1.12       | 2.36           | 3.45         | 4.90        |
| Perceived Partner Porn Use            | 4.50          | 2.22       | 1.42       | 2.70           | 3.56         | 4.89        |
| Aggressive Porn Alone                 | 1.59          | 1.31       | 1.05       | 2.20           | 3.33         | 4.73        |
| Aggressive Joint Porn Use             | 1.14          | 1.05       | 1.01       | 2.04           | 3.35         | 5.09        |
| Perceived Partner Aggressive Porn Use | 3.92          | 1.18       | 1.06       | 2.11           | 3.44         | 4.90        |
| Pornography Acceptance                | 3.22          | 3.62       | 2.57       | 3.54           | 3.49         | 3.93        |
| Typical Pornography Duration          | 2.34          | 2.66       | 1.18       | 2.92           | 3.26         | 3.60        |
| Masturbation to Porn                  | 50.91         | 75.72      | 5.80       | 58.27          | 57.98        | 63.78       |

six-profile model with some slight variations by gender, we elected to proceed with our group comparisons across profiles on the full sample using the six-profile solution while controlling for gender in subsequent analyses looking at well-being and relationship quality.

### 3.3. Group comparisons

We next sought to explore how profiles of pornography use may differ based on a variety of individual and demographic factors. The goal with these analyses was to explore how demographic and background factors may be associated with varying pornography usage patterns. First, a chi-square analysis was run to examine differences across pornography use profiles based on gender, given the prevalence of gender differences found in previous research (Carroll et al., 2017; Price et al., 2016). Profiles significantly differed based on gender ( $\chi^2(5) = 359.56, p < .001$ ). Females were overrepresented in the Partner User group (92.7% female) and the Abstainer profile (82.4% female). Men were more likely to be overrepresented in the Heavy User profile (75.0% male). Other profiles offered a generally even split by gender.

A chi-square analysis was also run to examine differences across pornography use profiles based on relationship status. This was significant ( $\chi^2(25) = 133.29, p < .001$ ). An examination of cell frequencies suggested that those in the Partner Use profile were less likely to be married (only 40.1%) compared to those in other profiles and were also more likely to be in a cohabiting relationship (31.6%) compared to those in other profiles. Those in the Heavy Use profile were slightly more likely to be in an exclusive dating relationship (15.7%) than those in other profiles who typically reported around 10% or less in this category.

Next, two MANOVA models were run with pornography use profile being the main factor of interest. The first MANOVA explored background demographic factors commonly related to each other, including age, religiosity, sexual orientation, and education. This overall model was significant (Wilk's  $\Lambda = 0.912, p < .001$ , partial  $\eta^2 = 0.023$ ). An exploration of univariate F-tests suggested that age ( $F(5, 3520) = 27.98, p < .001$ , partial  $\eta^2 = 0.038$ ), education ( $F(5, 3520) = 17.51, p < .001$ , partial  $\eta^2 = 0.024$ ), sexual orientation ( $F(5, 3520) = 2.61, p = .023$ , partial  $\eta^2 = 0.004$ ), and religiosity ( $F(5, 3520) = 23.05, p < .001$ , partial  $\eta^2 = 0.032$ ) were all related to significant differences across pornography profiles. Follow-up post-hoc tests controlling for multiple comparisons via a Bonferroni adjustment suggested that the significant findings for sexual orientation resulted in no significant pairwise comparisons, suggesting this result was likely spurious. When considering age, those in the Abstainer profile were significantly older than those in each of the other profiles. Those in the Solo User profile were also significantly younger than those in the Strong User profile. In terms of education, those in the Partner Porn User profile reported significantly less education than those in each of the other profiles. Additionally, those in the Strong User and Heavy User profiles reported significantly more education than those in the Abstainer and Solo User profiles, while those in the Heavy User profile also reported significantly more education than those in the Moderate User profile. Finally, when examining religiosity, those in the Abstainer profile were significantly more

religious than those in each of the other profiles except those in the Strong User profile and Heavy User profile. Additionally, and paradoxically, those in the Heavy User profile reported the highest level of religiosity and were significantly higher than those in each of the other profiles. Those in the Strong User group reported significantly higher religiosity than those in the Partner and Solo user profiles.

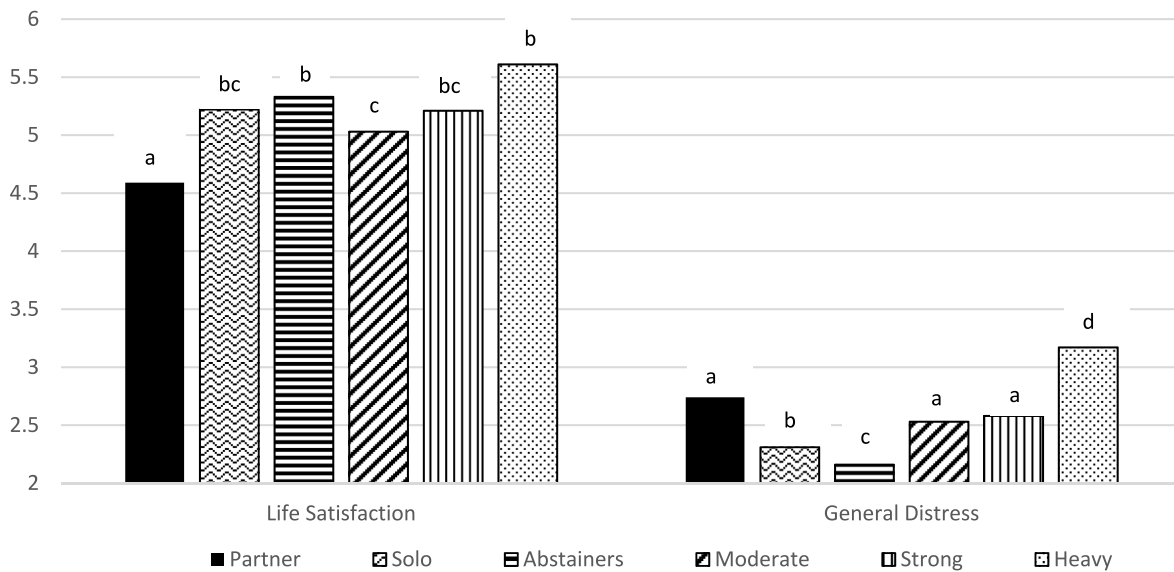
We next explored differences based on two correlated family and relational background factors (number of biological children and relationship length) by examining a second MANOVA model. This overall model was significant (Wilk's  $\Lambda = .958, p < .001$ , partial  $\eta^2 = 0.021$ ). An exploration of univariate F-tests suggested that number of biological children ( $F(5, 3383) = 5.37, p < .001$ ; partial  $\eta^2 = 0.008$ ) and relationship length ( $F(5, 3383) = 25.41, p < .001$ , partial  $\eta^2 = 0.036$ ) both resulted in significant differences across profiles. Follow-up post-hoc tests controlling for multiple comparison via a Bonferroni adjustment suggested that the significant finding for biological children resulted from differences between the Solo User profile and those in the Partner Porn Use and Abstainer profile. Those in the Solo User profile reported significantly less children than these other two profiles. Additionally, those in the Heavy User profile reported significantly less children than those in the Partner User profile. In terms of relationship length, those in the Abstainer profile reported significantly longer-term relationships than those in other profiles. Those in the Solo User profile reported significantly longer relationships than each of the other profiles except those in the Abstainer and Partner Use profile.

### 3.4. Differences in well-being and relationship quality

Our final analyses examined potential differences across pornography use profiles on individual well-being and relationship quality indicators. We again ran two models based on correlated dependent variables, one focused on individual well-being with general distress and life satisfaction used in the first MANCOVA model. The second MANCOVA model explored relationship quality, with relationship satisfaction and stability as the main dependent variables. Both models controlled for religiosity, biological children, gender, education, relationship status, relationship length, and age, all factors found to differ by profile based on previous results.

For the model examining individual well-being, results suggested that pornography profile was significantly related to differences across individual well-being (Wilk's  $\Lambda = .93, F(10, 6446) = 23.16, p < .001$ , partial  $\eta^2 = 0.035$ ). Univariate F-tests found that significant differences across pornography profile were found for both life satisfaction ( $F(5, 3224) = 10.22, p < .001$ , partial  $\eta^2 = 0.016$ ) and distress ( $F(5, 3224) = 32.27, p < .001$ , partial  $\eta^2 = 0.048$ ). Fig. 1 shows a graphical representation of adjusted mean differences based on post-hoc tests controlling for multiple comparisons with a Bonferroni adjustment while specific estimated means for these results can be found in Table 3. These results found that for life satisfaction, those in the Partner Use profile reported significantly less life satisfaction than those in all other profiles. Those in the Moderate Use profile also reported significantly lower life satisfaction than those in both the Heavy User and Abstainer profiles.

For general distress, stronger differences were found. Those in the



**Fig. 1.** Mean Differences Between Pornography User Profiles on Individual Well-being

Note. Profiles with the same letter denotes non-significant differences ( $p < .05$ ). All analyses controlled for number of biological children, education, relationship status, age, relationship length, gender, and religiosity.

**Table 3**

Adjusted mean differences between pornography use profiles on individual and relational well-being.

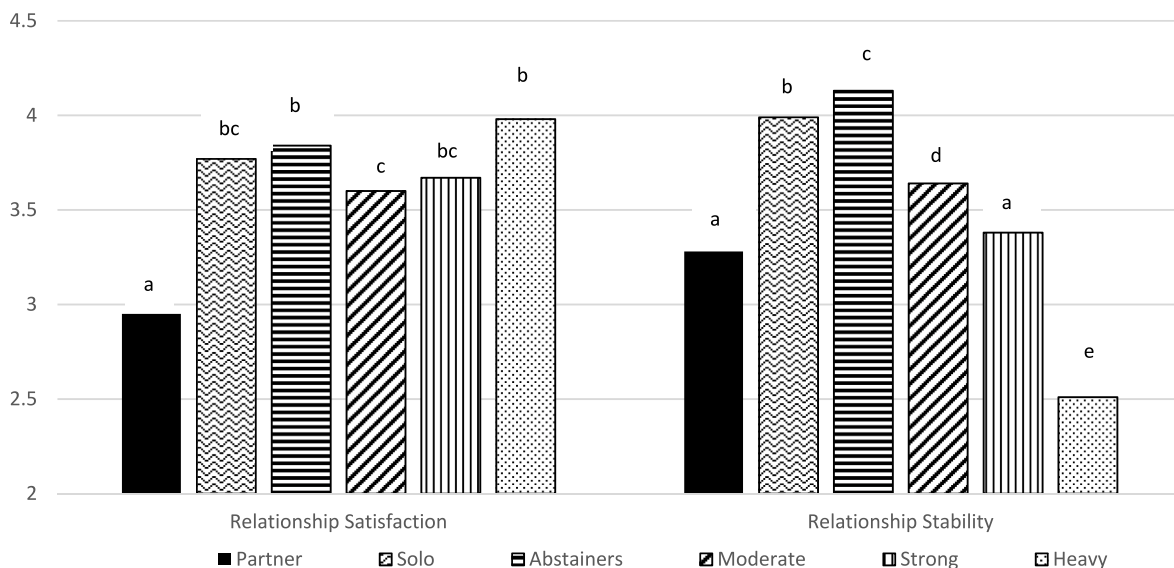
| Variable                  | Partner Users     | Solo Users         | Abstainers        | Moderate Users    | Strong Users       | Heavy Users       |
|---------------------------|-------------------|--------------------|-------------------|-------------------|--------------------|-------------------|
| Life Satisfaction         | 4.59 <sup>a</sup> | 5.22 <sup>bc</sup> | 5.33 <sup>b</sup> | 5.03 <sup>c</sup> | 5.21 <sup>bc</sup> | 5.61 <sup>b</sup> |
| General Distress          | 2.74 <sup>a</sup> | 2.31 <sup>b</sup>  | 2.16 <sup>c</sup> | 2.53 <sup>a</sup> | 3.67 <sup>a</sup>  | 3.17 <sup>d</sup> |
| Relationship Satisfaction | 2.96 <sup>a</sup> | 3.77 <sup>bc</sup> | 3.84 <sup>b</sup> | 3.60 <sup>c</sup> | 3.67 <sup>bc</sup> | 3.98 <sup>d</sup> |
| Relationship Stability    | 3.28 <sup>a</sup> | 3.99 <sup>b</sup>  | 4.13 <sup>c</sup> | 3.64 <sup>d</sup> | 3.38 <sup>ad</sup> | 2.51 <sup>e</sup> |

Note. Models controlled for biological children, education, relationship status, age, relationship length, gender, and religiosity.

Differing superscripts represent significant differences ( $p < .05$ ).

Heavy Use profile reported the highest levels of distress and were significantly different than each other profile. Those in Abstainer profile reported significantly lower distress than those in all other profiles. Those in the other profiles reported means between these two profiles, with most groups being significantly different from each other. These results provided general support for H1.

For the model examining relational quality, the overall model was significant (Wilk's  $\Lambda = 0.82$ ,  $F(10, 6516) = 68.02$ ,  $p < .001$ , partial  $\eta^2 = 0.095$ ) indicating the significant differences existed across pornography use profiles on relationship quality indicators. Step-down univariate F-tests were then examined for both relationship satisfaction and stability. Both the model for satisfaction ( $F(5, 3259) = 22.66$ ,  $p < .001$ , partial  $\eta^2$



**Fig. 2.** Mean Differences Between Pornography User Profiles on Relationship Well-being

Note. Profiles with the same letter denotes non-significant differences ( $p < .05$ ). All analyses controlled for relationship length, gender, and religiosity.



= 0.034) and stability ( $F(5, 3259) = 72.85, p < .001$ , partial  $\eta^2 = 0.126$ ) were significant, suggesting that for both factors, significant differences existed across pornography use profiles. Post-hoc pairwise comparisons were examined that utilized a Bonferroni adjustment to control for multiple comparisons. Estimated means for these results can be found in Table 3 and are graphically displayed in Fig. 2. As can be seen, for satisfaction, those in the Partner Porn User profile reported significantly less relationship satisfaction than those in all other profiles. Those in the Moderate Use profile also reported significantly lower satisfaction than those in both the Abstainer and Heavy Use profile.

For stability, results largely mirrored those of individual distress. Those in the Abstainer profile reported significantly higher stability than those in all other profiles. Those in the Heavy Use profile reported significantly less stability than those in all other profiles. Those in the other profiles reported means between these two profiles with most groups being significantly different from each other. This provided general support for hypothesis 2.

#### 4. Discussion

Results from the present study provide important new insights into the study of pornography. Findings confirm that pornography use is likely insufficiently captured by simple frequency measures that assess only a continuum from no use to heavy or compulsive use. As suggested by previous studies who have also found distinct groups of pornography users (Brown, Conner, & Vennum, 2017; Brown, Durtschi, et al., 2017; B  the et al., 2020; St  lhofer et al., 2019), simply using a single frequency measure to capture pornography use likely misses important between-person differences in how individuals (and couples) commonly approach pornography. Latent profile results suggested the presence of at least six distinct types of pornography users. While three of these profiles (Moderate Users, Strong Users, and Heavy Users) fell along this commonly assumed continuum and largely mirror similar groups found in previous studies, more robust measurement led to the discovery of three additional profiles that represented unique behavioral and attitudinal approaches to individual and couple pornography use. These profiles differed across almost all variables explored and provide important avenues for future research to explore.

One notable exception was that few differences across profiles were found between mainstream and aggressive pornography use. These content areas appeared to largely trend together, suggesting that, at least in this dataset, the content of pornography viewed was a poor way to differentiate different patterns of pornography use. However, given that previous scholars have noted differences in viewing patterns based on specific themes (see Hald & St  lhofer, 2016), it is possible that attempting to group pornography viewing into two broad categories did not provide enough specificity in pornography themes to detect differences. A more nuanced measurement of content in future studies may be needed to differentiate pornography users based on the content they are viewing.

Profile results included a large group of individuals who fell within the Abstainer profile. Similar to patterns found among adolescent and young adult samples (Willoughby et al., 2018), this Abstainer profile was not compromised only of those who never engaged with pornography. Instead, this represented a large pool of individuals who infrequently utilized pornography alone or with a partner and tended to have neutral to negative views of pornography in general. While pornography may be becoming increasingly normative (Price et al., 2016), the presence of this large group of individuals who appear to largely and perhaps intentionally avoid pornography suggests that while complete avoidance of pornography is rare, regular behavioral patterns of pornography use noted in other studies which often suggest normative weekly use (Carroll et al., 2008) are perhaps heavily influenced by a smaller group of individuals who engage in pornography use on a consistent basis. As has been noted in previous studies of pornography frequency, most of these Abstainers were women (Carroll et al., 2008, 2017), providing

further evidence for long standing gender differences seen between men and women in their engage with pornographic material (see Willoughby, Rhoades, & Carroll, 2021).

The other two unique profiles found were for those in romantic partnerships where one individual reported regular pornography use in the absence of either joint pornography use or the pornography use of the other partner. Both the Partner Porn Use and Solo Use profiles represented different sides of the same coin. While those in the Partner Porn Use profile represented those partners who are aware of their partner's use but opt to not view pornography regularly on their own, the Solo User profile likely represents partners who engage with pornography alone with low frequency partners. It is worth noting that the primary reason for this use appeared to be connected to masturbation, something noted or assumed by other scholars (Miller et al., 2019; Perry, 2020). Both groups reported little to no joint pornography use. It is important to note that the Partner Porn User profile was predominantly female. While the Solo User profile was also mostly female (61%), this high concentration of women in the Partner Porn User profile suggests that not only do men and women tend to differ in their engagement with pornography generally, but that in many heterosexual relationships, women who abstain from pornography use may be disproportionately more likely than men to be in a relationship with a partner who regularly uses pornography on their own. This is also in line with research on couple pornography frequencies which has suggested that this pornography use discrepancy between genders, sometimes call the "porn gap" (Carroll et al., 2017), is common. These results both confirm and extend previous findings related to the dyadic nature of modern pornography use.

In terms of individual and relational outcomes, those in the Abstainer profile tended to report the best quality relationships and best individual well-being compared to those in other profiles. This is in line with other research which has suggested that pornography use may be linked (although often inconsistently) associated with self-defined negative individual (Hald & Malamuth, 2008; Miller et al., 2018) and relational (B  the et al., 2021; Daneback et al., 2009; Poulsen et al., 2013; Willoughby et al., 2016) outcomes, especially if such use is heavy or compulsive (Harper & Hodgins, 2016). While many moderating and contextual factors likely shift and change this relationship, the pattern of those generally avoiding pornography use reporting more positive well-being is in line with most previous research and provided evidence in support of both hypotheses 1 and 2.

It is possible that this pattern has less to do with the absence of pornography and more to do with the avoidance of a topic that often causes stress and conflict among romantic couples. Abstainers may have similar morals, values, or attitudes about pornography that helps them avoid a potentially contentious issue in their relationship. The directionality of this association is important to consider. While research focused on the acquisition and expression of sexual scripts has suggested that pornography use may result in shifts in sexual expectations and desires in ways that may undermine healthy relationship functioning (Wright et al., 2013; Wright, Herbenick, & Tokunaga, 2022), it is also possible that individuals who avoid pornography are also more likely to engage in healthy relationship behaviors, as evidenced by the fact that pornography use is sometimes associated with both personality factors (Egan & Parmar, 2013) and risk-taking behaviors (Harkness, Mullan, & Blaszczyński, 2015), that may also be related to negative well-being and couple processes.

Those in the Partner Porn User generally reported the lower outcomes than those in other profiles, reporting the lowest levels of both life satisfaction and relationship satisfaction while also reported low levels of stability and high levels of general distress. These findings perhaps relate to research suggesting that when couples are misaligned in their views and behaviors tied to pornography use, relationship quality and stability suffers (Carroll et al., 2017; Willoughby et al., 2016). These divergent pornography usage patterns between couples may represent deeper and more problematic value differences between partners that

can create conflict around both pornography use specifically and intimacy generally.

As noted, it is interesting that those in the Solo User profile would also be in romantic partnerships that at first glance would also represent couples with differing viewpoints on pornography. Yet those in this profile tended to report well-being at levels that did not differ from those in other profiles. Perhaps this may be explained by the gender differences between the profiles. Several scholars have noted that female partners in relationships often have negative reactions to their male partner's pornography use (Resch & Alderson, 2014; Tylka & Kroon Van Diest, 2015). Perhaps some of the difference in outcomes between these profiles is attributed to the fact that the Partner Porn User profile is almost entirely comprised of women. In addition, it is possible that pornography users who use pornography in isolation are unaware of the negative impact it is having on their relationship and are less likely to self-report negative outcomes. Future research should continue to consider the differences between individuals in the Solo Use and Partner Use profiles to better understand the dyadic process at play in these unique partnerships.

The Heavy User profile represented one of the more interesting groups in the present study. It should be stated that caution should be taken in interpreting the results from this profile given their highly unusual profile and small group size. While finding a profile that represented high engagement with pornography was expected, that this group would also be highly religious was not. While some limited research has hinted that religious individuals may be more prone toward compulsive pornography use (Borgogna, Isacco, & McDermott, 2020), we do not discount the possibility that this profile may also represent measurement error in the dataset. We did specifically explore if perhaps this profile represented individuals who simply picked high values on every item in the survey. This exploration did not yield compelling results as many scales in the dataset included reversed scored items and those in the Heavy User profile had response patterns on these items in line with expectations (marking scores in patterns expected on each scale). If taken at face value, this profile seems to represent a conflicted and small group of individuals who heavily engage in pornography despite their religious beliefs or who may hold strong religious beliefs tied to faiths that do not have strong moral opposition to pornography. While we were not able to assess the specifics of these religious beliefs and it is possible these individuals represent spiritual people who do not come from traditional conservative Christian faiths who generally oppose pornography, the well-being profile of this user profile does suggest the potential presence of moral incongruence typically associated with negative well-being (Grubbs & Perry, 2019). Their high pornography use in the presence of high religiosity may create cognitive dissonance at levels that undermined both relationship stability and increase personal distress.

Overall, the variety of findings and differences found across the six profiles suggested at minimum the need for researchers to explore the influences and correlates of pornography use with more caution and nuance. While significant associations between individual and relational quality and pornography use frequency were perhaps likely to be found at the simple bivariate level (and preliminary analyses suggested they were), simply arguing that pornography use has simple and linear associations with these outcomes may mask the complexity that exists when it comes to how pornography use is navigated at the individual and relational level. While we did find that those in the Abstainer profile were the most likely to report positive well-being, understanding that this was not simply the avoidance of mainstream pornography (the focus of most previous research) but the avoidance of all forms of pornographic material combined with more negative attitudes toward pornography is an important extension of previous findings. Combined with recent research suggesting that the links between pornography use and relational well-being are likely non-linear (Willoughby, Leonhardt, & Augustus, 2021), and scholars need to continue to observe and document this complexity in the scholarship moving forward.

This study has several limitations that should be considered. First, the results of this study should not be interpreted to mean that only six profiles of pornography users exist. The purpose of this study was to illustrate the potential between-person differences that likely exist among pornography users, not to definitively suggest a complete theoretical package of only six potential usage patterns. Given that the sample was overrepresented by women, it is possible that this slightly altered the configuration of the profiles (although our gender sensitivity analyses suggested any gender variation in profile results was minimal). Variations in measurement and the addition (or subtraction) of additional contextual factors would likely alter the number of profiles found. Another limitation is that we utilized only self-report measures for one person in each relationship. As such, we have no way to validate the actual pornography use of romantic partners. Individuals may be unaware of a partner's pornography use or may have inaccurate estimates of how much pornography their partner views. It is possible that if the specific partner use was known, profile patterns and outcomes would differ. Results here should be replicated in true dyadic data sets to see how well the profiles proposed here are replicated in couple datasets.

The sample from this study was also taken from a sample only from the United States. Patterns of pornography use may be different in other regions of the world. Cross-cultural research on pornography is currently rare and more complex patterns of pornography use should be explored in various countries to determine if cross-cultural differences exist. Effect sizes were generally in the small to moderate range, suggesting that links between pornography use and individual and relational well-being are likely modest. Pornography use is likely one of many factors that contribute to an individual's well-being and relational quality. Finally, while our measurement allowed us to explore differences based on mainstream and aggressive pornography content, this broad distinction did not allow us to examine additional differences that may emerge based on viewing specific pornography themes such as BDSM, orgies, or lesbian/gay content. Additional common pornography content areas were not captured by our measurement (mature, fetish, amateur). Given other research which has noted differences in viewing patterns based on specific themes (see Hald et al., 2018), future research should consider how more specific content may alter the profiles found here.

Results from the present study expand previous research on pornography use patterns among those in romantic partnerships while providing additional new avenues of scholarship moving forward. First, results suggest that simply relying on the basic frequency of individual mainstream pornography use as the primary marker of pornography use likely minimizes even common normative patterns of pornography use that likely exist. Results also suggest that these varying patterns are related to variations in both individual and relational well-being. Regardless of the directionality of these associations, their presence here and across previous studies continues to suggest the importance of scholars understanding how common patterns of normative pornography use are related to both individual and relational health.

#### Author credit statement

Brian Willoughby Conceptualization; Data curation; Formal analysis; Methodology; Project administration; Supervision; Writing – original draft; Writing – review & editing. Carson Dover Conceptualization; Data curation; Writing – original draft; Writing – review & editing. Jason Carroll Conceptualization; Writing – review & editing. Matthew Saxey Conceptualization; Writing – review & editing.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The authors do not have permission to share data.

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